

The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	AIE 3905
Course title (English)	AI and Society in Action I
Course title (Chinese)	AI与社会实践 I
Units	3
Description (English)	Centered on "social practice" as its core methodology, this course adopts a tripartite teaching model integrating lecture guidance, experimental validation, and field investigation. Guiding students to step out of the classroom and into real-world settings, the curriculum utilizes diverse approaches—including corporate visits, community research, policy analysis, and hands-on project implementation—to critically examine the practical efficacy and ethical boundaries of AI technologies across scenarios of industrial innovation, urban governance, and social development. The course is dedicated to cultivating students' technical insight, social responsibility, and cross-disciplinary collaborative capabilities. Throughout the semester, students will engage in a complete chain of responsibility, encompassing pre-action research, critical reflection during action, and post-action impact assessment, thereby achieving the integration of knowledge and practice.
Description (Chinese)	本课程以“社会实践”为核心方法论，采用“讲座引领、实验验证、实地考察”三位一体的教学模式。课程引导学生走出课堂、深入现场，通过企业参访、社区调研、政策分析及项目实战等多元化途径，深度剖析AI技术在产业创新、城市治理与社会发展中的应用实效与伦理边界。课程致力于培养学生的技术洞察力、社会责任感及跨领域协作能力。学生在这门课程中将完成行动前的调研、行动中的反思、行动后的影响评估这一完整责任链条。

B. Corresponding course at CUHK

Please give details of the *closest* corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	
Description (Chinese)	

There is no corresponding course at CUHK.

2. **Course requisites**

Please state prerequisites and co-requisites, in terms of courses in CUHK-Shenzhen* or any other requirements (e.g., having taken certain subjects in high school).

(* Because course codes may not yet be stable, please provide both course code and course title.)

A. **Prerequisites**

AIE 1001-1002 or CSC 1001-1002 or CSC 1003-1004 or CSC 1005-1006 (or equivalents demonstrating baseline programming literacy).

B. **Co-requisites**

N.A

C. **Others**

This course may be offered across multiple parallel sessions, each focusing on a distinct theme or specific societal context. Individual sessions may have their own session-specific prerequisites.

3. **Learning outcomes**

Upon successful completion of this course, students will be able to:

Knowledge Integration

- 1) Explain the fundamental principles, capabilities, and limitations of AI technologies (including LLMs, Agents, and automated workflows) in the context of real-world social and industrial issues.
- 2) Apply systems analysis frameworks (such as design thinking and stakeholder mapping) to identify and define authentic problems within a specific social or industry domain (e.g., urban governance, regional economical growth, environment and sustainable development, healthcare, culture heritage, rural revitalization).
- 3) Identify relevant policy environments², safety and compliance requirements, ethical norms, and diverse stakeholder interests that shape the application of AI

in society.

Skill Development

- 4) Collaborate effectively in cross-disciplinary teams by taking on defined professional roles (e.g., product/project manager, user researcher, AI technical lead) and communicating across different disciplinary perspectives.
- 5) Build functional AI solution prototypes (e.g., AI Agents, intelligent workflows, or digital SOPs) using low-code/no-code platforms, informed by fieldwork insights such as user interviews and pain point analysis.
- 6) Iterate project designs through an agile process, incorporating feedback gathered from real-world user testing in field settings, and present final outcomes through professional project pitching and public exhibition.

Personal Growth

- 7) Demonstrate a “technology-for-good” value orientation by proactively addressing AI ethics, data privacy, and social responsibility in project design and implementation decisions.
- 8) Develop psychological resilience and decision-making agility when working under conditions of ambiguity, uncertainty, and setbacks encountered during real-world problem-solving.
- 9) Articulate a reflective personal narrative that details how the course experience fostered intellectual curiosity, empathy for diverse users, and respect for disciplinary perspectives beyond one’s own.

4. Course syllabus

These topics are organized into thematic clusters to reflect the course’s interdisciplinary, project-based nature. No single session covers all topics above. Each parallel session selects and adapts topics based on its specific theme, partner base, student composition, and semester resources.

1) Foundational AI Concepts for Social Practice

- Basic principles and capabilities of large language models (LLMs)
- Introduction to AI Agents and autonomous workflows
- Prompt engineering for real-world problem-solving
- Low-code/no-code AI prototyping tools (e.g., workflow builders)
- Understanding AI’s limitations: bias, hallucination, and context windows

2) Social Science Research Methods for AI Projects

- Design thinking for AI-enabled solutions
- Stakeholder mapping and interest analysis
- Conducting semi-structured interviews in field settings
- Pain point identification
- User testing methods for AI prototypes in real environments

3) AI Ethics, Law, and Governance

- Algorithmic bias and fairness in social applications
- Data privacy, consent, and security in field research
- AI transparency and explainability for non-expert users
- Regulatory landscapes for AI in different sectors (e.g., healthcare, education)
- Ethical review processes for socially situated AI projects

4) Domain-Specific Applications (Illustrative)

These topics vary by session theme.

AI + Future Industry

- Human-AI collaboration in manufacturing workflows
- AIGC's impact on creative industries (cost reduction, copyright issues)
- AI-driven process optimization and digital SOP design

AI + Urban Governance

- Algorithmic accessibility for vulnerable groups (e.g., elderly, disabled)
- Balancing data privacy with public efficiency (e.g., smart communities)
- AI-assisted public service delivery and citizen feedback loops

AI + Healthcare

- AI-assisted triage and patient trust
- Ethical boundaries of medical LLMs in grassroots health education
- AI for health behavior change communication

AI + Educational Equity

- Personalized AI micro-course generation for under-resourced schools
- Algorithmic recommendation bias and adolescent cognitive development
- Low-infrastructure AI tools for remote or offline settings

AI + Rural Revitalization

- AI applicability in low-resource environments (limited connectivity, devices)
- Digital empowerment for aging populations in rural areas
- AI-supported agricultural extension and market access

AI + Cultural Heritage

- Digital human interaction for museum collections
- AIGC-assisted preservation and re-narration of intangible cultural heritage
- Cross-cultural digital storytelling and global outreach

5) Project Management & Team Dynamics

- Role-based collaboration in cross-disciplinary teams (PM, user researcher, AI

lead)

- 360° peer feedback and team contract design
- Agile iteration under field uncertainty and time constraints
- Problem reframing based on real-world user feedback

6) Communication & Impact Delivery

- Data visualization and storytelling for non-technical stakeholders
- Producing demo videos for AI prototypes
- Writing policy briefs or ethics audit reports with field evidence
- Public pitching and joint exhibitions across parallel sessions
- Social impact assessment for AI interventions

7) Reflection & Professional Growth

- Documenting “real-world complexity” in field reflection logs
- Analyzing personal experiences of leaving disciplinary comfort zones
- Building resilience through project setbacks and pivots
- Articulating a “technology-for-good” personal narrative

5. Assessment scheme

Component/ method	% weight
Participation, Labs & Field Engagement	30
Milestone deliverables	30
Final project deliverables	40

Penalty on unapproved absence: All absences, including those unexpected, should be notified to the instructor by email BEFORE class. Supporting documents, if any, should be provided within 2 days after the leave. In general, leaves without advanced approval would NOT be accepted. Unapproved absence will result in **immediate withdrawal** from the course.

6. Grade descriptor

Grade	Description
A	Demonstrates the ability to achieve all the learning outcomes in this course and be able to follow all "Best-Practice" in accordance to assessment and presentations. Students shows outstanding performance that surpass the normal expectation at this level with exceptional skills in creativity and innovation, both in class and take-home assignments
A-	Mostly follow "Best-Practice" in accordance to assessment and presentations with minor, non-critical errors found. Students demonstrates strong ability in understanding and applying all "Best-Practice" requirements in a clear and best manner
B	Generally follow "Good" requirements in accordance to assessment and presentations with several non-critical errors. Students show strong ability in both "Best-Practice" & "Good" areas with setbacks and has the ability to apply the knowledge in a satisfactory and unambiguous manner
C	Generally follow "Good/Average" requirements with several errors including critical ones. Students show general understanding of the principles learnt in the course in a

	manner that is not incorrect but is somewhat fragmented.
D	Generally follow "Average/Poor" requirements with several errors including critical ones. Students show basic understanding of the principles learnt in the course in a manner that is broadly correct in its essential in some simple and familiar situations.
F	Unsatisfactory performance on a number of learning outcomes; OR failure to meet majority of assessment requirements.

7. **Feedback for evaluation**

- Course and teaching evaluation survey will be conducted at the end of the course. Students are reminded of their responsibility and right to give feedback to facilitate enhancement of the course.
- Students are welcome to give feedback to the course teacher at any time in person or through emails.

8. **Readings**

A. **Required**

B. **Recommended**

9. **Course components**

Activity	Hours/week
Lecture , Lab/Workshop, or Field trip	3

10. **Indicative teaching plan**

This course embraces a flexible structure rather than a rigid week-by-week schedule. Instruction flows dynamically across three core phases anchored by four critical milestones, allowing maximum breathing room for project evolution and teaching teams.

Phase 1: Team Formation & Problem Locking (Week 1-4)

- Phase Objective: Align cross-disciplinary team cognition, establish team charters, connect with partner bases, and lock onto a verified, real-world problem.
- Instructional Actions:
 - In-Class (Guidance/Validation): Universal no-code AI toolkit training (LLM applications, Agent building, Prompt Engineering); sociological fieldwork methodologies and ethics training.
 - Out-of-Class (Fieldwork Warm-up): Cross-disciplinary teaming and role assignment (e.g., Project Manager PM, User Researcher UR, Technical Officer TE); review the demand lists released by partner bases and conduct primary literature research.
- Critical Milestone 1 (Around Week 3-4): **Proposal Defense & Ethical Review**
 - Delivery Requirements: Submit a Project Design Proposal (Clearly answering:

Where are you going? Whose pain point are you solving? What is the initial AI intervention concept? How will you mitigate data and ethical risks?). Teams can only enter the field after passing this review.

Phase 2: Scenario Immersion & Agile Iteration (Week 5-10)

- Phase Objective: Move frequently in and out of the field to validate problems, avoid "armchair philosophy," rapidly incubate solution prototypes using AI tools, and test them in the real world.
- Instructional Actions:
 - In-Class (Guidance/Validation): Guest lectures from course instructor or industry/base experts; prototype design workflow guidance; peer-review sessions.
 - Out-of-Class (Deep Fieldwork): Enter the "Fieldwork Intensive Period." Teams embed themselves in bases for interviews and observation. Guided by validation mentors, they use low-code platforms or design Standard Operating Procedures (SOPs) to build an AI-powered solution prototype. Prototypes must be brought back to the field for real-user testing.
- Critical Milestone 2 (Around week 7): Midterm Review & Presentation
 - Delivery Requirements: Present a User Pain-Point Analysis Canvas and Version 1.0 AI Solution Prototype (Focusing on evaluating whether the pain point is real and whether the technical path is feasible, with industry mentors participating in the critique).
- Critical Milestone 3 (Late Phase 2): Field User Testing Feedback Loop
 - Delivery Requirements: Submit a Real-User Testing Report (How did the solution perform in a real scenario? What feedback was gathered? How has the solution been iterated based on that feedback?).

Phase 3: Synthesis & Giving Back (Week 11-14)

- Phase Objective: Translate projects into high-quality deliverables, give back to the bases and industry, and complete a profound personal "growth narrative."
- Instructional Actions:
 - In-Class (Guidance/Validation): Social impact assessment methodologies; project roadshow storytelling and data visualization guidance.
 - Out-of-Class (Public Voice): Finalize project packaging (videos, posters, or live system demos); officially present the final solutions or consultation briefs back to the partner base/enterprise.
- Critical Milestone 4 (End of Semester): Final Exhibition & Fireside Review
 - Delivery Requirements: Multi-class joint public roadshow exhibition (evaluated by a combined panel of academic judges and base mentors) + Team Final Project Report + Individual Growth Reflection Logs.

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections.

Expected Enrolment: Planned for up to 10 parallel experiential sections university-wide, capped at a maximum of 30 active student practitioners per section to ensure intensive, high-fidelity community interaction, product validation mentoring, and field guidance.

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all fieldwork logs, laboratory code, venture reports, and product presentations with professional integrity. Given the explicit thematic orientation of this course, the use of generative AI utilities is permitted—and encouraged—as a catalyst for collaborative ideation, product prototyping, qualitative coding support, and multi-lingual translation. However, unauthorized or unacknowledged automated generation passed off as independent primary field research or unverified code synthesis constitutes serious academic dishonesty.

Misuse includes fabricating stakeholder interviews via LLMs, deploying unvalidated code snippets in simulated community contexts without risk warnings, or failing to transparently log prompt histories in project code repositories. Every final submission must include an operational codebase alongside a thorough AI Disclosure Appendix outlining tools utilized, query strategies deployed, and human-supervised validation steps executed.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences?

Submit application

Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

Yes

14. Any other information

15. Version date

Version number	1
As of (date)	2026-06-16